

HDOC DAY-15 REPORT

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Activity	HDOC Day-15
Topic	Menu-driven C program using switch-case

Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice: (1) check whether all digits are even, or (2) check whether all digits are prime.

1. Algorithm

1. Start.
2. Input a positive integer.
3. If the number is less than or equal to 0, display a message asking for a positive integer and stop.
4. Display the menu: (1) Check whether all digits are even, (2) Check whether all digits are prime.
5. Read the user choice and copy the input number into a temporary variable.
6. For choice 1, extract each digit using remainder by 10. If any digit is odd, mark the condition false; otherwise continue until all digits are checked.
7. Display either "All digits are even" or "All digits are not even".
8. For choice 2, extract each digit. A digit is prime only if it is 2, 3, 5, or 7. If any digit is different, mark the condition false.
9. Display either "All digits are prime" or "All digits are not prime".
10. If the choice is neither 1 nor 2, display "Invalid choice".
11. Stop.

2. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Number: 12345 Choice: 1	All digits are not even	All digits are not even	Pass
2	Number: 24680 Choice: 1	All digits are even	All digits are even	Pass
3	Number: 2357 Choice: 2	All digits are prime	All digits are prime	Pass
4	Number: 2379 Choice: 2	All digits are not prime	All digits are not prime	Pass

3. Program Code

The following screenshots show the C program written for the Day-15 activity.

```
UW PICO 5.09 File: day15.cpp

#include <stdio.h>

int main()
{
    long long number, n;
    int choice, digit, flag = 1;

    printf("Enter a positive integer: ");
    scanf("%lld", &number);

    if (number <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Check whether all digits are even\n");
    printf("2. Check whether all digits are prime\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    n = number;

    switch (choice)
    {
        case 1:
            while (n > 0)
            {
                digit = n % 10;

                if (digit % 2 != 0)
                {
                    flag = 0;
                    break;
                }

                n = n / 10;
            }

            if (flag == 1)
                printf("All digits are even\n");
            else
                printf("All digits are not even\n");

            break;

        case 2:
            while (n > 0)
            {
                digit = n % 10;

                if (!(digit == 2 || digit == 3 ||
                    digit == 5 || digit == 7))
                {
                    flag = 0;
                    break;
                }

                n = n / 10;
            }

            if (flag == 1)
                printf("All digits are prime\n");
            else
                printf("All digits are not prime\n");

            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

```
UW PICO 5.09 File: day15.cpp

    {
        digit = n % 10;

        if (digit % 2 != 0)
        {
            flag = 0;
            break;
        }

        n = n / 10;
    }

    if (flag == 1)
        printf("All digits are even\n");
    else
        printf("All digits are not even\n");

    break;

    case 2:
        while (n > 0)
        {
            digit = n % 10;

            if (!(digit == 2 || digit == 3 ||
                digit == 5 || digit == 7))
            {
                flag = 0;
                break;
            }

            n = n / 10;
        }

        if (flag == 1)
            printf("All digits are prime\n");
        else
            printf("All digits are not prime\n");

        break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

4. Compilation, Execution and Testing

The program was compiled and executed in the terminal. The output verifies the prepared test cases for both menu choices.

```
Last login: Mon Sep  7 20:36:57 on ttys000
om@OMs-MacBook-Air ~ % nano day15.cpp
om@OMs-MacBook-Air ~ % clang day15.cpp -o day15
om@OMs-MacBook-Air ~ % ./day15
Enter a positive integer: 12345

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are not even
om@OMs-MacBook-Air ~ % ./day15
Enter a positive integer: 24680

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even
om@OMs-MacBook-Air ~ % ./day15
Enter a positive integer: 2357

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime
om@OMs-MacBook-Air ~ % █
```

Conclusion: The menu-driven program successfully uses switch-case to check whether all digits of a positive integer are even or prime. The tested outputs match the expected results.